OpenGeoSys Training 2026 – KIGAM

Lecture 3: Groundwater Modeling

Olaf Kolditz, Mehran Ghasabeh

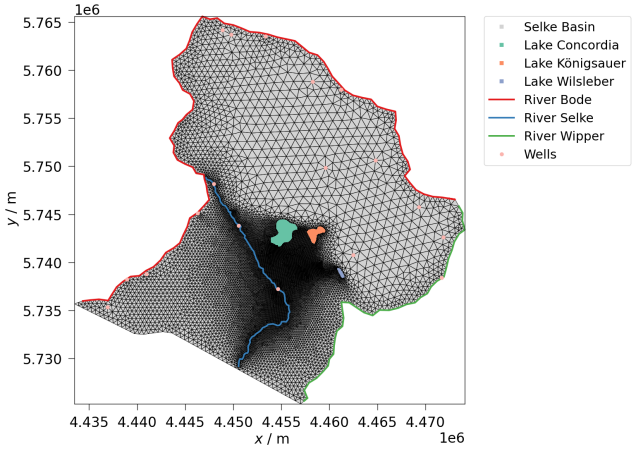
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Daejeon, 11th October 2026

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Agenda

1. Selke Catchment
2. Model Definition
   - Geometry
   - Meshing
   - ToDo (after OGS CM)
   - OGS Project
3. Model Run
4. Model Analysis
5. References



Lecture 3

Groundwater Modeling

Erik Nixdorf, Niklas Ritter, Tobias Meisel, Linus Walter, Thomas Kalbacher, Olaf Kolditz

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Groundwater modelling

Selke catchment (Zhang et al. 2021)

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AGU ADVANCING EARTH AND SPACE SCIENCE

Water Resources Research

RESEARCH ARTICLE

10.1029/2020WR029079

Key Points:

- Across a wide range of hydrologic scenarios, at least 34% of the stream length is characterized by losing conditions
- Heterogeneous solute concentrations in groundwater induce a different solute source composition from stream water source composition
- Most of the groundwater contributions to stream water are generated over only a small fraction of total stream length

Supporting Information:

Supporting Information may be found in the online version of this article.

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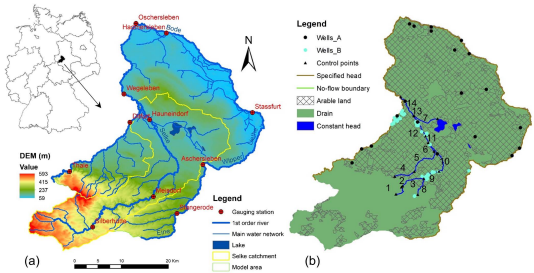
Effects of Heterogeneous Stream-Groundwater Exchange on the Source Composition of Stream Discharge and Solute Load

Zhi-Yuan Zhang[1], Christian Schmidt[1,2], Erik Nixdorf[3], Xingxing Kuang[4], and Jan H. Fleckenstein[3,5]

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Abstract The exchange of water and solutes between the stream and groundwater along a stream network affects the source composition of discharge and solute load in the stream. To date, this hydrologic turnover has only been analyzed with respect to the exchange of water. In this study, we extend the concept of hydrologic turnover to solutes and analyze the effects of different hydrologic conditions on the spatial patterns and magnitude of exchange fluxes. Based on a coupled stream-groundwater model built in MODFLOW using the Streamflow-routing package, we simulated stream-groundwater exchange along a stream of 30 km in length and evaluated the evolution of stream water source composition under different precipitation and streamflow conditions. Results show that even for highly variable hydrologic conditions, the direction of stream-groundwater exchange (loss/gain) remained unchanged in consistently gaining or losing reaches, but changed in interspersed transitional reaches. The comparison between the source composition of discharge and nitrate load in stream water revealed a decoupling of discharge and solute load contribution from groundwater to the stream, as zones of high water gains do not necessarily coincide with zones of high solute concentrations. Overall, nearly 80% of total groundwater contributions to the outflow of water from the catchment were generated over only 20% of the total stream length. Our research highlights the importance of distinguishing water and solute load contributions to streams. This in turn implies that to reduce groundwater-borne nutrient loads to streams, measures need to focus on the specific reaches with highest load contributions.

Plain Language Summary Streams and groundwater are connected and continuously exchange water and solutes. For example, the inflow of groundwater into streams not only increases streamflow, but also influences the chemical composition of stream water if the groundwater contains different solutes than the stream. Therefore understanding the spatial patterns of water and solute contributions from groundwater to the stream is important for the management of stream water quality. Taking the lower Selke River in central Germany as an example, we investigated the influence of stream gains and losses on the source composition of discharge and solute loads along the stream network as well as the influence of different precipitation and streamflow conditions. We found that higher precipitation and the associated elevated groundwater recharge or lower stream flows can increase the total length of gaining reaches, decreasing the relative importance of the source water from the upper stream. Due to the



Groundwater modelling

Selke catchment (Nixdorf, Sips and Morstein 2025)

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Routledge
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TECHNICAL REPORT OPEN ACCESS Check for updates

Development of a digital workflow for layer reduction strategies and visual-analytical outcome analysis to enhance geo-hydraulic modelling efficiency

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ABSTRACT

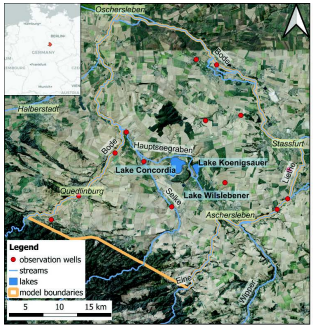
Introduction: In numerical groundwater flow modelling, construction and parametrisation of complex 3D stratigraphic systems is resource-intensive, notably elevating computational requirements. Simplifying the hydrogeological system assists in overcoming these bottlenecks, but preserving the underlying physical properties and providing simulation results with an acceptable deviation from the original model remains a challenge.

Objective: This study presents a combined approach, leveraging expert knowledge to simplify stratigraphy and analyse changes in geohydraulic simulation outputs based on selected strategies.

Methodology: We develop and merge two open source tools: "Stratmerge" for calculating simplified stratigraphy extents and parameters as well as "Stacked Parameter Relation" for visual data exploration of geohydraulic ensemble simulations. We test the suitability of our approach by simulating groundwater flow in the Selke River Basin, Germany. The underlying hydrogeological model consists of eight layers of the former lignite mining area of Nachterstedt, Germany. The impact of each simplification scheme is evaluated by simulating groundwater flow with the software OpenGeoSys 6.4.

Results: We demonstrate that our combined approach is feasible and generic, simplifies model construction, and provides insight into the dependencies between the hydrogeological model setup and the simulation results. Our approach could be used by hydrogeological modellers for a wide variety of aquifers constituted of sedimentary granular material.

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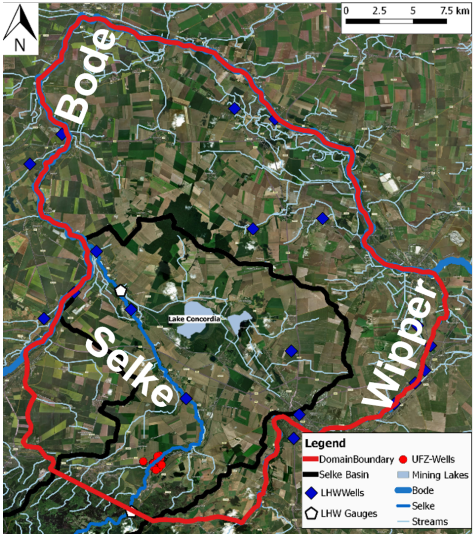


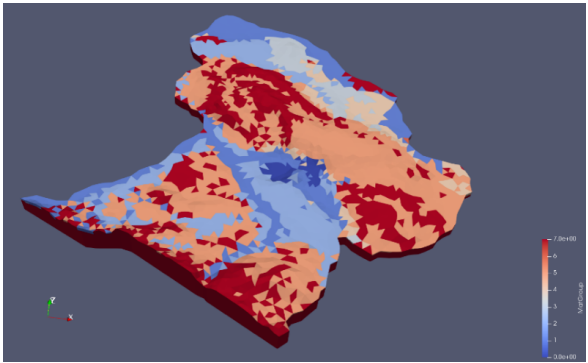
- Combined approach, leveraging expert knowledge to simplify stratigraphy and analyse changes in geohydraulic simulation outputs
- Workflow of open-source tools: Stratmerge (stratification) and simulator OGS

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Groundwater modelling

Selke catchment (Nixdorf, Sips and Morstein 2025) and Hydrosystems lecture at TU Dresden



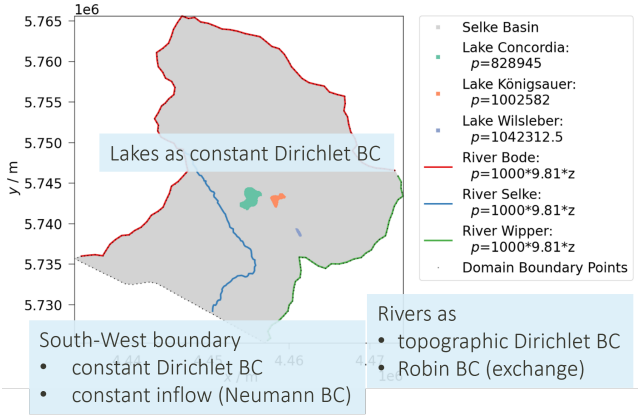


- Definition of boundary conditions (rivers, catchment, groundwater recharge)
- Aquifer properties (distribution)
- ⇒ Hydraulic influence of former open pit mining

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Groundwater modelling

Selke catchment - Model set-up, Model test run



Model geometry definition:

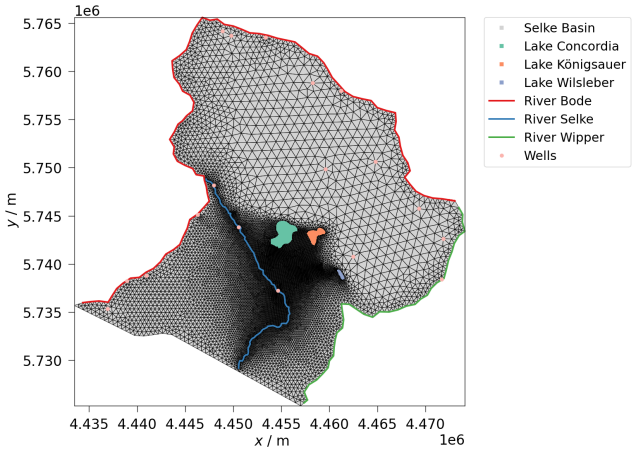
- Model domain: GIS > polyline for meshing
- Boundary conditions: extracting polylines from boundary mesh
- Meshes for lakes

Model test run:

- Process selection: simple (no couplings)
- Material properties: homogeneous
- Boundary conditions: Dirichlet (prescribed values)

Groundwater modelling

Selke catchment - OGS workflow, Jupyter notebooks



Notebooks overview for different boundary conditions:

- Dirichlet: prescribed hydraulic heads: $p = \rho g h = \rho g z$
- Neumann: groundwater fluxes
- Robin: river exchange

Notebooks intention:

- Reading existing OGS project
- Create an OGS project (todo)

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Groundwater modelling: Hackathon

Selke catchment - Model set-up #1: Geometries, meshes

ToDos (after OGS CM)

1. Model domain boundary: GIS-shp > gmsh polyline
2. Meshing: triangulation with gmsh
3. Mapping: mesh with DEM
4. River: including river polyline inside the model, remeshing
5. Lakes: including lake polygones inside the model, remeshing
6. Mapping: mesh with DEM

Results are required meshes (vtu) for the OGS project



Groundwater modelling

Selke catchment - Model set-up #1: OGS project

ToDos (after OGS CM)

1. Generate OGS H process template
2. Include meshes (see Model set-up #1: Geometries, meshes)
3. Assign boundary conditions: meshes
4. Assign boundary conditions: values, functions

Result is the complete OGS project



References

[1] Olaf Kolditz. Computational Methods in Environmental Fluid Mechanics. Springer Berlin Heidelberg, 2002. ISBN: 9783662047613. DOI: 10.1007/978-3-662-04761-3. URL: http://dx.doi.org/10.1007/978-3-662-04761-3.

[2] Erik Nixdorf, Mike Sips and Peter Morstein. 'Development of a digital workflow for layer reduction strategies and visual-analytical outcome analysis to enhance geo-hydraulic modelling efficiency'. In: Digital Water 3.1 (2025), pp. 1–24. DOI: 10.1080/28375807.2025.2460825.

[3] Charles V. Theis. 'The Relation Between the Lowering of the Piezometric Surface and the Rate and Duration of Discharge of a Well Using Ground-Water Storage'. In: Transactions, American Geophysical Union 16.2 (1935), pp. 519–524. DOI: 10.1029/TR016i002p00519.

[4] Zhi-Yuan Zhang, Christian Schmidt, Erik Nixdorf, Xingxing Kuang and Jan H. Fleckenstein. 'Effects of Heterogeneous Stream-Groundwater Exchange on the Source Composition of Stream Discharge and Solute Load'. In: Water Resources Research 57.8 (Aug. 2021), e2020WR029079. DOI: 10.1029/2020WR029079.